Grover's Algorithm

Assessment of quantum search

QUESTION PAPER

Time allowed: 3 hours **Maximum marks:** 240
Questions: 80 (4 sections × 20) **Answer key:** separate Answer Book

Name of the candidate: _____ **Roll No.:** _____

General instructions

1. This question paper has **four sections** of **20 questions** each: **A** Mathematics and Search Foundations, **B** Quantum Computing Foundations, **C** Grover's Algorithm Fundamentals, **D** Advanced Grover and Implementation.
2. Every question is of the **single-correct** type: four options, exactly one correct. Question formats used: single-correct MCQ, matching lists (List-I/List-II), Assertion–Reason, Statement I/II and paragraph-based questions.
3. Mark your answers on the **Response Sheet** at the end of this paper by darkening ONE bubble per question.
4. Calculators are **not** allowed. Rough work may be done on separate sheets.
5. Unless a question says otherwise, use the **conventions** printed below; they apply to every section.
6. The detailed solutions are in the separate *Answer Book*; do not consult it before finishing the test.

Marking scheme (every question)

Each question has **four** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Full marks: +3 if ONLY the correct option is chosen;
Zero marks: 0 if none of the options is chosen (question unanswered);
Negative marks: −1 in all other cases (wrong option, or more than one option chosen).

Conventions and notation used throughout the paper

- n qubits, $N = 2^n$ items, M marked items ($M = 1$ unless stated otherwise). A basis state is written $|x\rangle = |x_1 x_2 \dots x_n\rangle$: **qubit 1 is the most significant bit** and is written first. (The accompanying notebook writes `N_QUBITS` for n and $M = 2^N$ for the number of items, and O for the oracle.)
- $|s\rangle = H^{\otimes n}|0\dots0\rangle = \frac{1}{\sqrt{N}}\sum_x |x\rangle$. The oracle is $U_\omega|x\rangle = (-1)^{f(x)}|x\rangle$ with $f(x) = 1$ exactly on the marked items; the diffusion is $D = 2|s\rangle\langle s| - I$; one iteration is $G = DU_\omega$ (oracle first).
- $\sin\theta = \sqrt{M/N}$ with $0 < \theta \le \frac{\pi}{2}$; $|m\rangle$ and $|r\rangle$ are the uniform superpositions of the marked and the unmarked items. Then $G^k|s\rangle = \sin\big((2k+1)\theta\big)|m\rangle + \cos\big((2k+1)\theta\big)|r\rangle$, so $P_k = \sin^2\big((2k+1)\theta\big)$ and $k_{\text{opt}} = \text{round}\left(\frac{\pi}{4\theta} - \frac{1}{2}\right)$.
- **Textbook circuit:** $H^{\otimes n}$, then k times: [X on each qubit where x^* has a 0; a multi-controlled Z on all n qubits; the same X gates] (no ancillas) followed by [$H^{\otimes n}$; $X^{\otimes n}$; a multi-controlled Z; $X^{\otimes n}$; $H^{\otimes n}$]; finally measure all qubits.
- $H = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$; X, Y, Z are the Pauli matrices; $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. Two-qubit matrices use the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.
- All measurements are in the computational basis unless stated otherwise; log means $\log_2$. A numerical value given in a question may be used without a calculator.

SECTION A · Mathematics and Search Foundations Q.1–20

Syllabus: complex numbers, vectors and normalisation, inner products, trigonometric identities, reflections and rotations, inversion about the mean, Boolean counting, probability and classical search complexity.
Formats: Single-correct MCQ (Q.1–14); Matching lists (Q.15–16); Assertion–Reason (Q.17); Statement I / II (Q.18); Paragraph based (Q.19–20). Each question: +3 / 0 / −1.

Q.1 If $\sin\theta = \frac{1}{4}$, the value of $\sin^2 3\theta$ is

(A) $\frac{1}{16}$ **(B)** $\frac{121}{256}$ **(C)** $\frac{9}{16}$ **(D)** $\frac{11}{16}$

Q.2 Let $u = \left(\frac{3}{5}, \frac{4}{5}\right)^T$. The matrix $2uu^T - I$ equals

(A) $\frac{1}{25}\begin{pmatrix} 18 & 24 \\ 24 & 32 \end{pmatrix}$ **(B)** $\frac{1}{25}\begin{pmatrix} -7 & 12 \\ 12 & 7 \end{pmatrix}$ **(C)** $\frac{1}{25}\begin{pmatrix} -7 & 24 \\ 24 & 7 \end{pmatrix}$ **(D)** $\frac{1}{25}\begin{pmatrix} 7 & -24 \\ -24 & -7 \end{pmatrix}$

Q.3 A vector in the plane is reflected first in the line through the origin at angle 10° to the x-axis, and then in the line through the origin at angle 35°. The combined map is

(A) a reflection in the line at angle 45° **(B)** a rotation by $+25^\circ$
(C) a rotation by -50° **(D)** a rotation by $+50^\circ$

Q.4 Let $\sin\theta = \frac{1}{4}$ with $0 < \theta < \frac{\pi}{2}$, and let $G = \begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}$. The eigenvalues of G are

(A) $\frac{7 \pm \sqrt{15}}{8}$ **(B)** $\frac{\sqrt{15} \pm i}{4}$ **(C)** ± 1 **(D)** $\frac{7 \pm i\sqrt{15}}{8}$

Q.5 Let $z = \frac{1+i}{\sqrt{2}}$. The value of z^6 is

(A) $-i$ **(B)** -1 **(C)** 1 **(D)** i

Q.6 Let $\sin\theta = \frac{1}{10}$, $0 < \theta < \frac{\pi}{2}$; numerically $\theta = 0.10017$. Among the integers $0 \le k \le 10$, the one that makes $\sin^2\left((2k+1)\theta\right)$ largest is

(A) 8 **(B)** 10 **(C)** 5 **(D)** 7

Q.7 The vector $v \in \mathbb{R}^8$ has seven entries equal to $+1$ and one entry equal to -1. Each entry v_j is replaced by $2\bar{v} - v_j$, where $\bar{v}$ is the mean of the entries. The new value of the entry that was -1 is

(A) $\frac{5}{2}$ **(B)** $\frac{3}{2}$ **(C)** $\frac{7}{4}$ **(D)** $\frac{1}{2}$

Q.8 A box holds 31 cards of which exactly 3 are marked. Cards are drawn one at a time at random without replacement until the first marked card appears. The expected number of draws is

(A) $\frac{31}{4}$ **(B)** $\frac{31}{3}$ **(C)** 8 **(D)** 16

Q.9 A key, which may be absent, is looked up in a list of 1023 distinct entries using classical comparisons. The worst-case numbers of comparisons are, for binary search on the *sorted* list and for linear search on the *unsorted* list, respectively,

(A) 10 and 1023 **(B)** 1023 and 1023 **(C)** 10 and 512 **(D)** 32 and 1023

Q.10 Let $f(x_1, x_2, x_3, x_4) = (x_1 \vee x_2) \wedge (x_3 \oplus x_4)$ on $\{0,1\}^4$. The fraction of inputs with $f = 1$ is

(A) $\frac{1}{2}$ **(B)** $\frac{3}{8}$ **(C)** $\frac{1}{4}$ **(D)** $\frac{5}{8}$

Q.11 A Boolean function f on 4 bits takes the value 1 on exactly 3 inputs. The value of $\sum_{x \in \{0,1\}^4} (-1)^{f(x)}$ is

(A) 10 **(B)** 7 **(C)** 6 **(D)** 13

Q.12 The vector $|\psi\rangle = c\,(\,|00\rangle + i\,|01\rangle + (1+i)\,|10\rangle + 2\,|11\rangle\,)$ with $c > 0$ is normalised. The value of c is

(A) $\frac{1}{5}$ **(B)** $\frac{1}{2\sqrt{2}}$ **(C)** $\frac{1}{8}$ **(D)** $\frac{1}{\sqrt{7}}$

Q.13 Let $g(n) = \frac{\pi}{4}\sqrt{2^n}$. Then $g(n+6)/g(n)$ equals

(A) 6 **(B)** 8 **(C)** $\sqrt{6}$ **(D)** 64

Q.14 Let $u = \frac{1}{\sqrt{2}}(1, i, 0)^T$ and $v = \frac{1}{\sqrt{3}}(1, 1, 1)^T$. The inner product $\langle u|v\rangle$ is

(A) $\frac{1+i}{\sqrt{6}}$ **(B)** $\frac{2}{\sqrt{6}}$ **(C)** $\frac{1-i}{\sqrt{6}}$ **(D)** 0

Q.15 Match the value of $\sin^2\theta$ (List-I) with the value of $\sin^2 3\theta$ (List-II).

List-I		**List-II**	
(P)	$\sin^2\theta = \frac{1}{8}$	**(1)**	1
(Q)	$\sin^2\theta = \frac{1}{4}$	**(2)**	0
(R)	$\sin^2\theta = \frac{1}{2}$	**(3)**	$\frac{25}{32}$
(S)	$\sin^2\theta = \frac{3}{4}$	**(4)**	$\frac{1}{2}$

(A) P→4, Q→1, R→3, S→2 **(B)** P→3, Q→1, R→2, S→4
(C) P→3, Q→1, R→4, S→2 **(D)** P→1, Q→3, R→4, S→2

Q.16 Let $|u\rangle$ be a unit vector in $\mathbb{C}^N$, with $N \geq 4$. Match each matrix in List-I with its trace in List-II.

List-I		**List-II**	
(P)	$2\,|u\rangle\langle u| - I$	**(1)**	N
(Q)	$I - 2\,|u\rangle\langle u|$	**(2)**	$N-2$
(R)	$|u\rangle\langle u|$	**(3)**	1
(S)	I	**(4)**	$2-N$

(A) P→3, Q→2, R→4, S→1 **(B)** P→4, Q→2, R→1, S→3
(C) P→2, Q→4, R→3, S→1 **(D)** P→4, Q→2, R→3, S→1

Q.17 **Assertion (A):** Rounding $\frac{\pi}{4}\sqrt{N}$ to the nearest integer does not always give the integer k for which $(2k+1)\theta$ is closest to $\frac{\pi}{2}$, where $\sin\theta = 1/\sqrt{N}$.
Reason (R): For $0 < x < 1$ we have $\arcsin x < x$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Q.18 Let $|u\rangle$ be a unit vector in $\mathbb{C}^N$, with $N \geq 2$, and let $D = 2\,|u\rangle\langle u| - I$.
Statement I: $D^2 = I$.
Statement II: D has the eigenvalue $+1$ with multiplicity 1 and the eigenvalue -1 with multiplicity $N-1$.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Paragraph for Questions 19 and 20

Define $p(k) = \sin^2\left((2k+1)\theta\right)$ for integers $k \geq 0$, with $\theta = \pi/18 = 10^\circ$. (This is the shape of the Grover success curve, but the two questions below need only trigonometry.) Note that $(2k+1)\theta$ takes the values $10^\circ, 30^\circ, 50^\circ, \ldots$

Q.19 The smallest k with $p(k) = 1$, and the smallest positive period of the sequence $p(k)$, are

(A) first maximum at $k = 4$; smallest period 18

(B) first maximum at $k = 5$; smallest period 9

(C) first maximum at $k = 9$; smallest period 18

(D) first maximum at $k = 4$; smallest period 9

Q.20 The number of integers k with $0 \leq k \leq 17$ for which $p(k) \geq \frac{3}{4}$ is

(A) 3 **(B)** 6 **(C)** 9 **(D)** 4

SECTION B · Quantum Computing Foundations Q.21–40

Syllabus: superposition, registers and tensor products, Born rule, global and relative phases, Hadamard layers, Pauli and controlled gates, unitary operators, bit and phase oracles, kickback, entanglement and state-vector calculations.
Formats: Single-correct MCQ (Q.21–34); Matching lists (Q.35–36); Assertion–Reason (Q.37); Statement I / II (Q.38); Paragraph based (Q.39–40). Each question: +3 / 0 / −1.

Q.21 The state $H^{\otimes 3}\,|000\rangle$ is measured in the computational basis. The probability that qubit 1 gives 1 and qubit 3 gives 0 is

(A) $\frac{1}{8}$ **(B)** $\frac{1}{4}$ **(C)** $\frac{1}{\sqrt{8}}$ **(D)** $\frac{1}{2}$

Q.22 Let $|\psi\rangle = \frac{1}{\sqrt{2}}(\,|0\rangle + i\,|1\rangle\,) \otimes \frac{1}{\sqrt{2}}(\,|0\rangle - |1\rangle\,)$. The amplitude of $|10\rangle$ in $|\psi\rangle$ is

(A) $\frac{i}{2}$ **(B)** $-\frac{i}{2}$ **(C)** $\frac{1}{2}$ **(D)** $-\frac{1}{2}$

Q.23 $H \otimes H$ is applied to $\frac{1}{2}(\,|00\rangle + |01\rangle + |10\rangle - |11\rangle\,)$. The amplitude of $|11\rangle$ in the result is

(A) $\frac{1}{2}$ **(B)** 0 **(C)** $-\frac{1}{2}$ **(D)** 1

Q.24 Let $U_f\,|x\rangle\,|y\rangle = |x\rangle\,|y \oplus f(x)\rangle$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Then $U_f\,|x\rangle\,|-\rangle$ equals

(A) $|x\rangle\,|f(x)\rangle$ **(B)** $|x\rangle\,|-\rangle$ **(C)** $(-1)^{f(x)}\,|x\rangle\,|-\rangle$ **(D)** $|f(x)\rangle\,|-\rangle$

Q.25 With U_f as in the previous question and $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, the state $U_f\,|x\rangle\,|+\rangle$ equals

(A) $|x\rangle\,|-\rangle$ **(B)** $(-1)^{f(x)}\,|x\rangle\,|-\rangle$ **(C)** $(-1)^{f(x)}\,|x\rangle\,|+\rangle$ **(D)** $|x\rangle\,|+\rangle$

Q.26 Let CCZ be the three-qubit gate that multiplies only $|111\rangle$ by -1. Which basis state has its sign flipped by $X_2\,\mathrm{CCZ}\,X_2$, where X_2 is X on qubit 2?

(A) $|111\rangle$ **(B)** $|010\rangle$ **(C)** $|101\rangle$ **(D)** $|000\rangle$

Q.27 Let CCX be the Toffoli gate with target qubit 3. The product $(I \otimes I \otimes H)\,\mathrm{CCX}\,(I \otimes I \otimes H)$ equals

(A) CCX **(B)** $I \otimes I \otimes I$ **(C)** $Z \otimes Z \otimes Z$ **(D)** CCZ

Q.28 Which pair of states are physically *indistinguishable* (they give the same statistics for every possible measurement)?

(A) $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $-\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
(B) $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$
(C) $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
(D) $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $\frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$

Q.29 Let $|\psi\rangle = \frac{1}{\sqrt{10}}(\,|00\rangle + 2\,|01\rangle - 2\,|10\rangle + i\,|11\rangle\,)$. A measurement asks only whether the two qubits have *different* values (outcome "different" on the span of $|01\rangle, |10\rangle$). The probability of "different" and the state afterwards are

(A) $\frac{4}{5}$ and $|01\rangle$ **(B)** $\frac{4}{5}$ and $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$
(C) $\frac{4}{5}$ and $\frac{1}{\sqrt{10}}(2\,|01\rangle - 2\,|10\rangle)$ **(D)** $\frac{2}{5}$ and $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

Q.30 Let $|u\rangle$ be a unit vector in a space of dimension $N \geq 2$. Which of these operators is *not* unitary?

(A) $H^{\otimes n}$ (for $N = 2^n$) **(B)** $|u\rangle\langle u|$
(C) $I - 2\,|u\rangle\langle u|$ **(D)** $2\,|u\rangle\langle u| - I$

Q.31 For $N = 16$ let $|s\rangle$ be the uniform superposition of all 16 basis states and $|r\rangle$ the uniform superposition of 15 of them (all except one). The value of $\langle r|s\rangle$ is

(A) $\frac{\sqrt{15}}{4}$ **(B)** $\frac{15}{16}$ **(C)** $\frac{1}{4}$ **(D)** $\frac{1}{\sqrt{15}}$

Q.32 A state vector of $n = 33$ qubits is stored in double precision (16 bytes per complex amplitude). The memory needed is

(A) 256 GiB **(B)** 16 GiB **(C)** 64 GiB **(D)** 128 GiB

Q.33 For a single qubit ($n = 1$), the operator $2\,|+\rangle\langle+| - I$ equals

(A) X **(B)** H **(C)** Z **(D)** $-I$

Q.34 For n qubits let $|s\rangle = H^{\otimes n}\,|0\ldots0\rangle$ and $D = 2\,|s\rangle\langle s| - I$. If the oracle marks nothing (it acts as the identity), then after k iterations the state is

(A) $(-1)^k\,|s\rangle$ **(B)** $|0\ldots0\rangle$ for odd k
(C) a state whose amplitudes depend on k **(D)** $|s\rangle$ for every k

Q.35 Match each expression in List-I with its value in List-II.

List-I		**List-II**	
(P)	$H\,\|1\rangle$	**(1)**	$-\,\|-\rangle$
(Q)	$Z\,\|-\rangle$	**(2)**	$\|0\rangle$
(R)	$X\,\|-\rangle$	**(3)**	$\|+\rangle$
(S)	$H\,\|+\rangle$	**(4)**	$\|-\rangle$

(A) P→4, Q→1, R→3, S→2 **(B)** P→4, Q→3, R→2, S→1
(C) P→4, Q→3, R→1, S→2 **(D)** P→3, Q→4, R→1, S→2

Q.36 Match each two-qubit state in List-I with its description in List-II.

List-I		List-II	
(P)	$\frac{1}{2}(\|00\rangle + \|01\rangle + \|10\rangle + \|11\rangle)$	**(1)**	entangled, with four amplitudes of equal size
(Q)	$\frac{1}{2}(\|00\rangle - \|01\rangle - \|10\rangle + \|11\rangle)$	**(2)**	the product $\|-\rangle\|-\rangle$
(R)	$\frac{1}{2}(\|00\rangle + \|01\rangle + \|10\rangle - \|11\rangle)$	**(3)**	entangled, with only two non-zero amplitudes
(S)	$\frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle)$	**(4)**	the product $\|+\rangle\|+\rangle$

(A) P→4, Q→2, R→3, S→1 **(B)** P→4, Q→2, R→1, S→3
(C) P→4, Q→1, R→2, S→3 **(D)** P→2, Q→4, R→1, S→3

Q.37 **Assertion (A):** Measuring in the computational basis immediately after a phase oracle acts on $|s\rangle$ gives a uniformly random string; the oracle step alone reveals nothing.
Reason (R): The oracle multiplies each amplitude by ± 1, and the Born probabilities depend only on the magnitudes of the amplitudes.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Q.38 **Statement I:** The states $|\psi\rangle$ and $-|\psi\rangle$ give identical statistics for every measurement.
Statement II: Multiplying only *one* basis-state amplitude of a superposition by -1 can never change the statistics of any measurement.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Paragraph for Questions 39 and 40

Three qubits start in $|0\rangle|0\rangle|1\rangle$ (input qubits 1 and 2, ancilla qubit 3). A Hadamard gate is applied to all three. Then the Toffoli gate is applied with qubits 1 and 2 as controls and qubit 3 as target: it implements $U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$ with $f(x_1x_2) = x_1 \wedge x_2$.

Q.39 The state of the three qubits after the Toffoli gate is

(A) $\frac{1}{2}(|00\rangle - |01\rangle - |10\rangle - |11\rangle)|-\rangle$ **(B)** $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)|+\rangle$
(C) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)|-\rangle$ **(D)** $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)|-\rangle$

Q.40 Now the ancilla alone is measured in the computational basis. The probability of the outcome 1, and the state of the input qubits afterwards, are

(A) $\frac{1}{4}$; the input qubits collapse to $|11\rangle$
(B) $\frac{1}{2}$; the input qubits remain in $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$
(C) $\frac{1}{2}$; the input qubits collapse to $|11\rangle$
(D) 1; the input qubits remain in $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$

SECTION C · Grover's Algorithm Fundamentals Q.41–60

Syllabus: the angle θ, phase oracle and diffusion operator, inversion about the mean, success probability, optimal iterations, several solutions, over-rotation and gate counts.
Formats: Single-correct MCQ (Q.41–54); Matching lists (Q.55–56); Assertion–Reason (Q.57); Statement I / II (Q.58); Paragraph based (Q.59–60). Each question: +3 / 0 / −1.

Q.41 Grover's algorithm is run on $n = 2$ qubits ($N = 4$) with one marked item. After exactly one iteration, the probability of measuring the marked item is

(A) $\frac{1}{2}$ **(B)** $\frac{1}{4}$ **(C)** $\frac{3}{4}$ **(D)** 1

Q.42 For $N = 8$ with one marked item, the probability of measuring the marked item after one iteration is

(A) 1 **(B)** $\frac{25}{32}$ **(C)** $\frac{5}{8}$ **(D)** $\frac{9}{16}$

Q.43 For $n = 4$ qubits with one marked item ($\theta \approx 0.2527$ rad), the optimal number of iterations k_{opt} and the resulting success probability are

(A) $k_{\text{opt}} = 2$ and $P \approx 0.908$ **(B)** $k_{\text{opt}} = 4$ and $P \approx 0.582$
(C) $k_{\text{opt}} = 3$ and $P = 1$ **(D)** $k_{\text{opt}} = 3$ and $P \approx 0.961$

Q.44 For $n = 2$ ($N = 4$), the matrix of the diffusion operator $D = 2\,|s\rangle\langle s| - I$ has

(A) diagonal entries $+\frac{1}{2}$ and off-diagonal entries $-\frac{1}{2}$
(B) diagonal entries $-\frac{1}{2}$ and off-diagonal entries $+\frac{1}{2}$
(C) diagonal entries -1 and off-diagonal entries $+1$
(D) all entries equal to $\frac{1}{2}$

Q.45 An oracle for the marked string $|0101\rangle$ is built as: X gates, then the multi-controlled Z on all four qubits, then the same X gates. The X gates act on

(A) qubits 2 and 4 **(B)** no qubit **(C)** qubits 1 and 3 **(D)** all four qubits

Q.46 Use the circuit in the conventions box and count each multi-controlled Z as one gate. The number of gates in *one* Grover iteration for $n = 6$, $x^* = 010011$ is

(A) 32 **(B)** 30 **(C)** 26 **(D)** 38

Q.47 Let C be the multi-controlled Z on n qubits. The circuit $H^{\otimes n} X^{\otimes n}\, C\, X^{\otimes n} H^{\otimes n}$ implements

(A) $2\,|s\rangle\langle s| - I = D$ **(B)** $I - 2\,|s\rangle\langle s| = -D$
(C) $I - 2\,|0\ldots0\rangle\langle 0\ldots0|$ **(D)** $I - 2\,|1\ldots1\rangle\langle 1\ldots1|$

Q.48 For $N = 16$ with one marked item, the amplitude of each *unmarked* item after one iteration is

(A) $\frac{1}{4}$ **(B)** $\frac{11}{16}$ **(C)** $\frac{3}{16}$ **(D)** $\frac{7}{32}$

Q.49 Let $N = 16$ with $M = 4$ marked items. After one iteration the probability of measuring a marked item is

(A) $\frac{3}{4}$ **(B)** $\frac{25}{32}$ **(C)** 1 **(D)** $\frac{1}{4}$

Q.50 Let $N = 8$ with $M = 4$ marked items. The probability of a marked outcome after k iterations is

(A) 1 for even k and 0 for odd k **(B)** 0 for $k = 1$

(C) 1 for $k = 1$ **(D)** $\frac{1}{2}$ for every k

Q.51 Let $N = 8$ with $M = 6$ marked items. The probability of a marked outcome after one iteration is

(A) $\frac{3}{4}$ **(B)** $\frac{1}{4}$ **(C)** 1 **(D)** 0

Q.52 On 2^{15} items with one marked item, $k_{\text{opt}} = 142$. If $k = 284$ iterations are used instead, the probability of the marked item is about

(A) 3×10^{-6} **(B)** $\frac{1}{2}$ **(C)** $\frac{1}{32768}$ **(D)** 1

Q.53 For $n = 20$ qubits with one marked item, Grover needs 804 iterations. A classical random search needs on average about 2^{19} queries. The classical cost divided by the quantum cost is about

(A) 650 **(B)** 2 **(C)** 1000 **(D)** 20

Q.54 A Grover circuit with success probability $P = 0.9613$ is run for 2000 shots. The expected marked count and its binomial standard deviation are about

(A) 1923 ± 44 **(B)** 1923 ± 9 **(C)** 2000 ± 0 **(D)** 1923 ± 77

Q.55 Match the number of qubits (one marked item) in List-I with k_{opt} in List-II. Use $k_{\text{opt}} = \text{round}\left(\frac{\pi}{4\theta} - \frac{1}{2}\right)$.

List-I	**List-II**
(P) $n = 2$	**(1)** 12
(Q) $n = 6$	**(2)** 25
(R) $n = 8$	**(3)** 1
(S) $n = 10$	**(4)** 6

(A) P→3, Q→4, R→1, S→2 **(B)** P→3, Q→1, R→4, S→2

(C) P→3, Q→4, R→2, S→1 **(D)** P→4, Q→3, R→1, S→2

Q.56 Match each part of the Grover circuit in List-I with its effect in List-II.

List-I		**List-II**	
(P)	$H^{\otimes n}$ on $\lvert 0 \dots 0\rangle$	**(1)**	returns one bit string with its Born probability
(Q)	X gates on the 0 bits of x^*, multi-controlled Z, the same X gates	**(2)**	creates the uniform superposition $\lvert s\rangle$
(R)	$H^{\otimes n}X^{\otimes n}$, multi-controlled Z, $X^{\otimes n}H^{\otimes n}$	**(3)**	reflects about $\lvert s\rangle$, up to a global phase
(S)	measuring all qubits	**(4)**	flips the sign of $\lvert x^*\rangle$ only

(A) P→2, Q→4, R→3, S→1 **(B)** P→2, Q→4, R→1, S→3

(C) P→4, Q→2, R→3, S→1 **(D)** P→2, Q→3, R→4, S→1

Q.57 **Assertion (A):** The success probability of Grover's algorithm does not depend on which single item is marked.

Reason (R): The oracle is a diagonal matrix in the computational basis.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Q.58 **Statement I:** In Grover's algorithm, applying more iterations never lowers the success probability.

Statement II: For one marked item, the optimal number of iterations grows as $\sqrt{N}$.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Paragraph for Questions 59 and 60

Grover's algorithm runs on $n = 3$ qubits with the single marked item $x^* = |101\rangle$. It starts from the uniform state $|s\rangle$, in which every amplitude is $a = 1/\sqrt{8}$. One iteration is the oracle U_ω followed by the diffusion D. Trace the amplitudes by hand with inversion about the mean.

Q.59 After the first iteration the amplitudes are

(A) marked $\frac{3}{2}a$; each unmarked $\frac{1}{2}a$

(B) marked $\frac{7}{4}a$; each unmarked $\frac{3}{4}a$

(C) marked $\frac{5}{2}a$; each unmarked $\frac{1}{2}a$

(D) marked $\frac{5}{2}a$; each unmarked $\frac{3}{4}a$

Q.60 After the second iteration the probability of measuring $|101\rangle$ is

(A) $\dfrac{25}{32}$ **(B)** $\dfrac{11}{16}$ **(C)** $\dfrac{121}{128}$ **(D)** 1

SECTION D · Advanced Grover and Implementation Q.61–80

Syllabus: several solutions, unknown M, exact search, amplitude amplification, quantum counting, multi-controlled gates, the CUDA-Q notebook, simulation and shots, memory, noise, complexity, applications and limitations.
Formats: Single-correct MCQ (Q.61–74); Matching lists (Q.75–76); Assertion–Reason (Q.77); Statement I / II (Q.78); Paragraph based (Q.79–80). Each question: +3 / 0 / −1.

Q.61 For $N = 1024$ with $M = 16$ marked items, the optimal number of iterations is

(A) 25 **(B)** 6 **(C)** 5 **(D)** 8

Q.62 When M is unknown, the algorithm of Boyer, Brassard, Høyer and Tapp picks k uniformly from $\{0, \ldots, m-1\}$. The average success probability is at least $\frac{1}{4}$ once $m \geq 1/\sin 2\theta$. For $N = 2^{20}$ and $M = 1$ this threshold on m is about

(A) 512 **(B)** 1024 **(C)** 804 **(D)** 2^{20}

Q.63 To obtain success probability exactly 1, the angle is reduced from θ to θ' so that $(2k+1)\theta' = \pi/2$ with $k = \lceil \frac{\pi}{4\theta} - \frac{1}{2} \rceil$. For $N = 16$, $M = 1$ this requires

(A) $k = 3$ and $\theta' = \pi/12$ **(B)** $k = 4$ and $\theta' = \pi/18$
(C) $k = 2$ and $\theta' = \pi/10$ **(D)** $k = 3$ and $\theta' = \pi/14$

Q.64 Which statement about the query complexity of unstructured search is correct?

(A) Grover's algorithm is optimal only when N is a power of 2
(B) Quantum algorithms need only $O(\log N)$ queries, because $n = \log_2 N$ qubits hold all N items
(C) The lower bound $\Omega(\sqrt{N})$ holds only for classical algorithms
(D) Every quantum algorithm needs $\Omega(\sqrt{N})$ oracle queries, so Grover's algorithm is optimal up to a constant

Q.65 The notebook's full-circuit statistics for $n = 15$ and $k = 142$ report 4275 `h` gates, 5964 `x` gates and 284 multi-controlled Z gates. In one iteration the diffusion uses $2n$ `x` gates and the oracle uses two `x` gates for each 0 in x^*. The number of 0s in the marked string is

(A) 9 **(B)** 21 **(C)** 6 **(D)** 12

Q.66 Amplitude amplification boosts a procedure $\mathcal{A}$ that succeeds with probability $a = 0.01$ ($\theta \approx 0.10017$). At the optimal k, one uses $\mathcal{A}$ or $\mathcal{A}^{-1}$ in total

(A) 7 times **(B)** 15 times **(C)** 100 times **(D)** 8 times

Q.67 On the CPU state-vector backend the notebook estimates the memory as $2^n \times 16$ bytes for the simulator's state plus one equal host copy, because `RUN_ITERATION_SWEEP = True` holds two states at once. Its report for $n = 15$ is 1.05 MB. With the same settings and $n = 28$ the estimate is

(A) 2 GiB **(B)** 16 GiB **(C)** 4 GiB **(D)** 8 GiB

Q.68 A cipher with a 128-bit key is attacked by Grover's algorithm (one correct key, cost measured in iterations). The number of iterations is about

(A) $\frac{\pi}{4} \cdot 2^{64}$ **(B)** $\frac{\pi}{4} \cdot 2^{127}$ **(C)** 2^{127} **(D)** 128

Q.69 A simple noise model multiplies the signal by f per iteration: $P_k \approx f^k \sin^2((2k+1)\theta) + (1-f^k)/N$. For $n = 10$, $k = 25$, $f = 0.99$ (take $0.99^{25} \approx 0.78$), the success probability is about

(A) 0.99 **(B)** 0.50 **(C)** 0.78 **(D)** 0.25

Q.70 A multi-controlled Z with m controls can be built with $m-1$ clean ancillas, $2(m-1)$ Toffoli gates and one CZ. For the $n = 10$ Grover register ($m = 9$), one full iteration (oracle and diffusion) needs how many Toffoli gates?

(A) 32 **(B)** 36 **(C)** 16 **(D)** 18

Q.71 On a state-vector simulator the work grows as (iterations) $\times$ (amplitudes per gate) $\sim 2^{n/2} \cdot 2^n$. Adding 2 qubits multiplies the total work by about

(A) 16 **(B)** 8 **(C)** 4 **(D)** $2\sqrt{2}$

Q.72 The notebook's sweep computes

```
sweep_max = 2 * k_optimal + 1
stride = max(1, ceil(sweep_max / SWEEP_MAX_POINTS))
```

It records the point $k = 0$ and then evolves by `stride` iterations `while k_done + stride <= sweep_max`, recording one point each time. For $k_{\text{opt}} = 142$ and `SWEEP_MAX_POINTS = 60`, the number of recorded points is

(A) 57 **(B)** 285 **(C)** 58 **(D)** 60

Q.73 A notebook run has theoretical success probability $P = 0.99998683$ and records 2000 marked counts out of 2000 shots. For independent shots, the probability of seeing all 2000 shots succeed is about

(A) 0.99999 **(B)** 0.026 **(C)** 0.500 **(D)** 0.974

Q.74 Let $N = 8$, $M = 1$, and repeat a run with a fixed k until the marked item is found, counting only oracle queries in the Grover iterations. Then $P_1 = \frac{25}{32}$ and $P_2 = \frac{121}{128}$. The expected numbers of queries for $k = 1$ and $k = 2$ are about

(A) 1 and 2 **(B)** 2.12 and 1.28 **(C)** 1.28 and 2.12 **(D)** 0.78 and 1.89

Q.75 Match each task in List-I with its query complexity in List-II.

List-I		**List-II**	
(P)	Unstructured search, classical	**(1)**	$\Theta(N^{1/3})$
(Q)	Unstructured search, quantum	**(2)**	$\Theta(N^{2/3})$
(R)	Collision finding in a 2-to-1 function, quantum	**(3)**	$\Theta(N)$
(S)	Element distinctness, quantum	**(4)**	$\Theta(N^{1/2})$

(A) P→3, Q→4, R→1, S→2 **(B)** P→4, Q→3, R→1, S→2

(C) P→3, Q→1, R→4, S→2 **(D)** P→3, Q→4, R→2, S→1

Q.76 Match each method in List-I with its main idea in List-II.

List-I		**List-II**	
(P)	Amplitude amplification	**(1)**	uses randomly chosen k from a growing range when M is unknown
(Q)	Quantum counting	**(2)**	changes the reflection phases so that extra steps cannot over-rotate
(R)	BBHT search	**(3)**	replaces $H^{\otimes n}$ by any state-preparation procedure $\mathcal{A}$
(S)	Fixed-point search	**(4)**	applies phase estimation to G to find M

(A) P→3, Q→1, R→4, S→2
(B) P→3, Q→4, R→1, S→2
(C) P→3, Q→4, R→2, S→1
(D) P→4, Q→3, R→1, S→2

Q.77 **Assertion (A):** Grover's algorithm solves NP-complete problems in polynomial time.
Reason (R): For a search over 2^n candidates, Grover's algorithm uses about $\frac{\pi}{4}2^{n/2}$ iterations.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Q.78 **Statement I:** The BBHT algorithm finds a marked item with an expected $O(\sqrt{N/M})$ queries without knowing M in advance.

Statement II: If exactly half of the items are marked, ordinary Grover iterations leave the success probability at $\frac{1}{2}$.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Paragraph for Questions 79 and 80

Quantum counting. A search space of $N = 64$ items contains an unknown number M of marked items; in fact $M = 16$. The Grover operator G has eigenvalues $e^{\pm 2i\theta}$ on the plane containing $|s\rangle$, with $\sin^2\theta = M/N$. Phase estimation of G with $t = 3$ counting qubits is run on $|s\rangle$, which has equal weight on the two eigenvectors. The outcome $y \in \{0, \dots, 7\}$ gives the phase estimate $\varphi \approx y/8$, where $e^{2\pi i\varphi}$ is the eigenvalue. From this one sets $\tilde{\theta} = \pi y/8$ and $\tilde{M} = N\sin^2\tilde{\theta}$.

Q.79 The two most likely outcomes are

(A) $y = 2$ and $y = 6$ **(B)** $y = 1$ and $y = 7$ **(C)** $y = 1$ and $y = 2$ **(D)** $y = 0$ and $y = 4$

Q.80 If $y = 1$ is obtained, the estimate of M is

(A) $\tilde{M} = 32$ **(B)** $\tilde{M} = 8$ **(C)** $\tilde{M} \approx 9.4$ **(D)** $\tilde{M} = 16$

RESPONSE SHEET

Darken exactly ONE bubble per question. Leave all four bubbles empty to skip a question (0 marks).

Name: _____ **Roll No.:** _____

Section A	Section B	Section C	Section D
1 (A)(B)(C)(D)	21 (A)(B)(C)(D)	41 (A)(B)(C)(D)	61 (A)(B)(C)(D)
2 (A)(B)(C)(D)	22 (A)(B)(C)(D)	42 (A)(B)(C)(D)	62 (A)(B)(C)(D)
3 (A)(B)(C)(D)	23 (A)(B)(C)(D)	43 (A)(B)(C)(D)	63 (A)(B)(C)(D)
4 (A)(B)(C)(D)	24 (A)(B)(C)(D)	44 (A)(B)(C)(D)	64 (A)(B)(C)(D)
5 (A)(B)(C)(D)	25 (A)(B)(C)(D)	45 (A)(B)(C)(D)	65 (A)(B)(C)(D)
6 (A)(B)(C)(D)	26 (A)(B)(C)(D)	46 (A)(B)(C)(D)	66 (A)(B)(C)(D)
7 (A)(B)(C)(D)	27 (A)(B)(C)(D)	47 (A)(B)(C)(D)	67 (A)(B)(C)(D)
8 (A)(B)(C)(D)	28 (A)(B)(C)(D)	48 (A)(B)(C)(D)	68 (A)(B)(C)(D)
9 (A)(B)(C)(D)	29 (A)(B)(C)(D)	49 (A)(B)(C)(D)	69 (A)(B)(C)(D)
10 (A)(B)(C)(D)	30 (A)(B)(C)(D)	50 (A)(B)(C)(D)	70 (A)(B)(C)(D)
11 (A)(B)(C)(D)	31 (A)(B)(C)(D)	51 (A)(B)(C)(D)	71 (A)(B)(C)(D)
12 (A)(B)(C)(D)	32 (A)(B)(C)(D)	52 (A)(B)(C)(D)	72 (A)(B)(C)(D)
13 (A)(B)(C)(D)	33 (A)(B)(C)(D)	53 (A)(B)(C)(D)	73 (A)(B)(C)(D)
14 (A)(B)(C)(D)	34 (A)(B)(C)(D)	54 (A)(B)(C)(D)	74 (A)(B)(C)(D)
15 (A)(B)(C)(D)	35 (A)(B)(C)(D)	55 (A)(B)(C)(D)	75 (A)(B)(C)(D)
16 (A)(B)(C)(D)	36 (A)(B)(C)(D)	56 (A)(B)(C)(D)	76 (A)(B)(C)(D)
17 (A)(B)(C)(D)	37 (A)(B)(C)(D)	57 (A)(B)(C)(D)	77 (A)(B)(C)(D)
18 (A)(B)(C)(D)	38 (A)(B)(C)(D)	58 (A)(B)(C)(D)	78 (A)(B)(C)(D)
19 (A)(B)(C)(D)	39 (A)(B)(C)(D)	59 (A)(B)(C)(D)	79 (A)(B)(C)(D)
20 (A)(B)(C)(D)	40 (A)(B)(C)(D)	60 (A)(B)(C)(D)	80 (A)(B)(C)(D)

Score computation (filled in by the evaluator, using the Answer Book)

Sec.	Topic	Attempted	Correct C	Wrong W	Marks $3C - W$
A	Mathematics and Search Foundations				
B	Quantum Computing Foundations				
C	Grover's Algorithm Fundamentals				
D	Advanced Grover and Implementation				
				Total (out of 240)	